AUTONOMOUS WAREHOUSE ROBOT TASK MANAGEMENT SYSTEM

1. Introduction

An Autonomous Warehouse Robot Task Management System is a software project that simulates how robots perform tasks in a modern warehouse. Warehouse robots may collect products, transport goods, and deliver packages to specified locations. When several tasks arrive, they must be organized and processed systematically. In this project, a queue stores pending tasks in First In, First Out (FIFO) order, while a stack stores completed task history so the most recently completed task can be viewed first. The menu-driven C++ program allows users to add tasks, process tasks, view pending work, and inspect or remove the latest history entry.

2. Problem Statement

A warehouse may receive many robot tasks at the same time, such as picking products, moving boxes, and delivering packages. Without an organized system, pending tasks can become difficult to track. A software system is therefore required to maintain a queue of pending tasks, process tasks in arrival order, and store completed-task history using a stack.

3. Objectives

- Manage warehouse robot tasks systematically.
- Use a queue to store and process pending tasks in FIFO order.
- Use a stack to maintain completed-task history in LIFO order.
- Display pending tasks and completed tasks.
- Handle empty queue and stack conditions safely.
- Demonstrate practical applications of stack and queue in automation.

4. Data Structures Used

Queue (FIFO): The first task inserted is the first task processed. Main operations are enqueue (add), dequeue (process/remove), and display.

Stack (LIFO): The latest completed task is at the top. Main operations are push (store), pop (remove the latest history entry), and display.

5. Main Features

- Add a new warehouse task.
- Process the oldest pending task.
- Display all pending tasks.
- Display completed task history, newest first.
- Remove the latest history entry.
- Exit through a menu-driven interface.